

LA-6: Data Visualization (15 points)

Learning Outcomes

In this assignment, you will:

- Learn how to use `ggplot2`, which is a package included in the `tidyverse` suite of packages, to create visualizations using data.
- Practice using functions from LA-5, including `mutate()`, `case_when()`, `freq()`, and `descr()`.
- Combine conditions with `|` (“or”).

Tip

Read all the instructions carefully before starting the assignment. Set up your R script as you did in LA-4 and LA-5 (name, uNID, date, assignment number).

Important

How to report your answers. Write each answer as a comment in your R script, labeled with its question number (e.g., `# 3a: ...`). Round averages to two decimal places and include units (e.g., \$180.38).

Instructions

- 1) Load the packages below. Then download the `Utilities.csv` dataset and its codebook from Canvas, upload the dataset to your Posit Cloud project (as you did in LA-4 and LA-5), and read it into R. Use the codebook to familiarize yourself with the data in this file.

- `tidyverse`
- `summarytools`

Note

About this dataset. Each row (case) is one monthly utility bill for **one household**, from 1999 to 2010. The bills include gas and electricity charges, how much gas and electricity the household used, and whether the household added a donation to *Operation Fuel* to that month’s bill.

- 2) Read and follow along with [Chapter 5 of this online book](#). Use the data provided in the chapter to follow the steps in Ch. 5.

💡 Tip

Do not skip Step 2 (above). It will help you understand `ggplot2`. You will use the same kinds of graphs from Ch. 5 below: a histogram (`geom_histogram()`), scatterplots (`geom_point()`), and a bar chart (`geom_col()`).

- 3) Make a histogram to display the distribution of the household's total monthly bill (`totalbill`).
 - a) Use the histogram you created and describe the distribution. Is the distribution skewed or symmetric? If skewed, is the distribution skewed positively or negatively?
 - b) Where do most of the total monthly bills fall? Give a range in dollars (e.g., "between \$X and \$Y").

💡 Tip

When you run `geom_histogram()`, R prints a message saying ``stat_bin()` using `bins = 30``. This is not an error. If you want to control the width of the bars, you can use, for example, `geom_histogram(binwidth = 25)` to make each bar cover \$25.

- 4) Make a scatterplot to display gas bill (`gasbill`) by month (`month`), with month on the x-axis. Read the tip below before you start, then use your graph to answer the following questions.
 - a) Looking at your graph, which month appears to have the highest gas bills?
 - b) Use R to calculate the average gas bill for the months of January and December. Which is higher?

💡 Tip

`month` is stored as a number (1 = January, 12 = December), so by default the x-axis is labeled 2.5, 5, 7.5, 10. To label every month, add `scale_x_continuous(breaks = 1:12)` to your plot:

df |>

```
ggplot(aes(x = month, y = variable)) +  
  geom_point() +  
  scale_x_continuous(breaks = 1:12)
```

For (b), use `filter()` and `descr()` as you did in LA-5, and read the **Mean** row.

- 5) Make a scatterplot to display electric bill (`elecbill`) by month (`month`), with month on the x-axis. Use your graph to answer the following questions.
 - a) Looking at your graph, which month appears to have the highest electric bills?
 - b) Use R to calculate the average electric bill for the months of September and December. Which is higher?
- 6) Make a scatterplot to display the relationship between gas usage (`ccf`) and gas bill (`gasbill`). Place gas usage on the x-axis and monthly gas bill on the y-axis. Does there appear to be a relationship between gas usage and gas bill? If so, describe the relationship (e.g., does a relationship exist? Is it linear? Is there a positive or negative relationship between the variables?).
- 7) Make a scatterplot to show the relationship between electricity usage (`kwh`) and electric bill (`elecbill`). Place electricity usage on the x-axis and monthly electric bill on the y-axis. Does there appear to be a relationship between electricity usage and electric bill? If so, describe the relationship (e.g., does a relationship exist? Is it linear? Is there a positive or negative relationship between the variables?).
- 8) Create a new categorical variable called `season`. Have the variable equal `winter` if the bill was from December, January, or February. Have the variable equal `spring` if the bill was from March, April, or May. Have the variable equal `summer` if the bill was from June, July, or August. Have the variable equal `fall` if the bill was from September, October, or November.

Then, examine a frequency table of `season`. Read the note below before you start.

- a) How many bills in the data set are from fall?
- b) How many bills in the data set are from summer?

Note

Use `mutate()` and `case_when()` the same way you did in LA-5. This time, each season covers three months, so you need to combine conditions with `|`, which means “**or**.” A row matches the condition if *any* part of it is TRUE. For example:

```
util <- util |>
  mutate(season = case_when(
    month == 12 | month == 1 | month == 2 ~ "winter",
    ...
  ))
```

Fill in the ... with a line for each of the other three seasons. Remember to save the result back to `util` (start with `util <- util |> ...`), because you will use `season` again in #9.

(In LA-5, you used `&`, which means “**and**”: a row matches only if *every* part is TRUE. A bill can’t be from December *and* January, so `&` won’t work here.)

- 9) **BONUS (2 points):** Make a bar chart using `geom_col()` to display the number of bills that included a donation in each `season` (the variable you created in #8).

First, make a new variable, `DonorStatus`. Set `DonorStatus` equal to 1 if the household donated money to *Operation Fuel* on that bill (`donate` is “yes”) and 0 otherwise. Check that you have created `DonorStatus` correctly by running `freq()` on both `donate` and `DonorStatus`: the number of **yes** bills should match the number of 1s, and the number of **no** bills should match the number of 0s.

Then, use `ggplot()` to create your graph, with `season` on the x-axis and `DonorStatus` on the y-axis. In your own words, describe the graph. Does donation vary by season?

Tip

`geom_col()` adds up the y values for each category on the x-axis. Because `DonorStatus` is 1 for every bill with a donation and 0 otherwise, adding up `DonorStatus` within a season gives the **number of bills with a donation** in that season.

Submission

Submit your R script (named `LA-#_FirstName-LastName.R`) to Canvas.

Your R script should:

- 1) Include commands and functions that are necessary to address all the questions in the assignment.
- 2) Contain comments that answer the questions in the assignment.
- 3) Run in its entirety without errors.

To ensure that your R script runs without errors, you should:

- Save your script.
- Navigate back to Your Workspace on Posit Cloud.
- Reopen your project.
- Run the entire script line-by-line without editing it to ensure there are no errors.

! Important

These standards apply to all submissions in this course that require R scripts. You should follow these instructions for preparation, naming, and saving of your R script for all of your individual lab assignments.